

Homework #5 - Spaceflight Mechanics

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1 Part 1

1.1 Problem

Design a 3-stage launcher intended to carry 2000 kg into a circular polar orbit of height 700 km. The required ΔV is estimated to be 9300 m/s. The first two stages use solid propellant with $I_{sp,1} = 268$ s and $I_{sp,2} = 285$ s respectively. For structural efficiency, values between 9 and 11 are assumed.

The third stage uses liquid propellant. Choose between the two propellants here below.

- NTO/MMH

$$I_{sp} = 320 \text{ s}$$

$$\epsilon = 6 \div 7$$

- LOX/LCH4

$$I_{sp} = 350 \text{ s}$$

$$\epsilon = 5 \div 6$$

1.2 Some definitions

Let's see some useful definitions for the launcher. These apply to each and every stage as well as the whole rocket.

$k = \frac{m_0}{m_f}$	mass ratio
$\epsilon = \frac{m_p + m_s}{m_s}$	structural efficiency
$\lambda = \frac{m_u}{m_0}$	payload ratio
$\xi = \frac{m_p}{m_0} = 1 - \frac{1}{k}$	propellant ratio

Also, this formula is very useful from time to time, we'll see it later.

$$\Rightarrow \frac{1}{k} = \lambda + \frac{1}{\epsilon}(1 - \lambda)$$

1.3 Preliminary design

We have chosen to use NTO/MNH as liquid propellant for the third stage. The other values were also chosen randomly in the given ranges. Down below there's a short summary of the situation.

stage	propellant	specific impulse I_{sp}	structural efficiency ϵ
1st stage	solid	268 s	9
2nd stage	solid	285 s	9.7
3rd stage	liquid	320 s	7

TABLE 1. Preliminary design of the rocket

We hypothesize $\bar{k} = \frac{m_0}{m_f} = \text{const}$ for all stages. This is only an assumption that we'll allow us to get a grasp on the situation and design a preliminary project.

$$\Delta V = \sum \Delta V_i = g_0 \ln k \cdot \sum I_{sp,i} = 9300 \text{ m/s}$$

$$\bar{k} = \exp\left(\frac{\Delta V}{g_0 \sum I_{sp,i}}\right) = 2.9622$$

Therefore for each stage we can estimate the ΔV produced.

$$\Delta V_i = g_0 I_{sp,i} \ln \bar{k}$$

$$\Delta V_1 = 2855 \text{ m/s}$$

$$\Delta V_2 = 3036 \text{ m/s}$$

$$\Delta V_3 = 3409 \text{ m/s}$$

To estimate the mass of each and every stage, and how it is redistributed throughout propellant, structural and payload, we need to start from the final payload and proceed backwards. Once we're done with the last stage, by repeating the same steps two more times we can obtain at the end the characteristics of the rocket launcher.

$$\Rightarrow m_{u,3} = 2000 \text{ kg}$$

$$\lambda_3 = \frac{\frac{\epsilon_3}{k} - 1}{\epsilon_3 - 1} = 0.2272$$

$$m_{0,3} = \frac{m_{u,3}}{\lambda_3} = 8803.3 \text{ kg}$$

$$m_{f,3} = \frac{m_{0,3}}{k} = 2971.9 \text{ kg}$$

$$m_{p,3} = m_{0,3} - m_{f,3} = 5831 \text{ kg}$$

$$m_{s,3} = m_{0,3} - m_{u,3} - m_{p,3} = 971.9 \text{ kg}$$

$$m_{tot,3} = m_{p,3} + m_{s,3} = 6803.3 \text{ kg}$$

$$\Rightarrow m_{u,2} = 8803.3 \text{ kg}$$

$$\lambda_2 = \frac{\frac{\epsilon_2}{k} - 1}{\epsilon_2 - 1} = 0.2615$$

$$m_{0,2} = \frac{m_{u,2}}{\lambda_2} = 33671 \text{ kg}$$

$$m_{f,2} = \frac{m_{0,2}}{k} = 11367 \text{ kg}$$

$$m_{p,2} = m_{0,2} - m_{f,2} = 22304 \text{ kg}$$

$$m_{s,2} = m_{0,2} - m_{u,2} - m_{p,2} = 2564 \text{ kg}$$

$$m_{tot,2} = m_{p,2} + m_{s,2} = 24868 \text{ kg}$$

$$\Rightarrow m_{u,1} = 8803.3 \text{ kg}$$

$$\lambda_1 = \frac{\frac{\epsilon_1}{k} - 1}{\epsilon_1 - 1} = 0.2548$$

$$m_{0,1} = \frac{m_{u,1}}{\lambda_1} = 132150 \text{ kg}$$

$$m_{f,1} = \frac{m_{0,1}}{k} = 44613 \text{ kg}$$

$$m_{p,1} = m_{0,1} - m_{f,1} = 87539 \text{ kg}$$

$$m_{s,1} = m_{0,1} - m_{u,1} - m_{p,1} = 10942 \text{ kg}$$

$$m_{tot,1} = m_{p,1} + m_{s,1} = 98482 \text{ kg}$$

We have finally obtained the preliminary design that guarantees a payload of 2000 kg for a mission that requires 9300 m/s of ΔV . FIG. 1 shows the characteristics.

LV Preliminary Design					
Stage	1	2	3	Payload	%
Inert	10942	2564	972	2000	0.0151
Prop	87539	22304	5831		LV(excl PL
Total	98482	24868	6803	2000	130153
Tot/in	9.00	9.70	7.00		
Initial	132153	33671	8803	2000	
Final	44613	11367	2972		
Isp	268.0	285.0	320.0		873.0
DeltaV	2855	3036	3409		9300
Kappa	2.962	2.962	2.962		
Lambda	0.255	0.261	0.227		

FIGURE 1. Initial configuration

2 Part 2

2.1 Problem

Optimize the previous design.

2.2 The constrained optimization

It's possible to optimize the previous design by using the Langrange multipliers method. It's the problem of constrained optimization. The first equation down below is the one to optimize, as a large Λ means an higher payload ratio, while the second equation is the constraint on the cost ΔV . We can rewrite both equations in such a way to show the characteristics of the launcher.

$$\begin{cases} \phi' = \ln \Lambda \\ \Delta V = \sum g_0 I_{sp,i} \ln k_i \end{cases}$$

$$\begin{aligned} \phi' &= \ln \Lambda \\ &= \ln \prod \lambda_i \\ &= \sum \ln \lambda_i \\ &= \sum \ln \left(\frac{\frac{\epsilon_i}{k_i} - 1}{\epsilon_i - 1} \right) \\ &= \sum \ln \left(\frac{\epsilon_i - k_i}{k_i(\epsilon_i - 1)} \right) \\ &= \sum [\ln(\epsilon_i - k_i) - \ln(k_i(\epsilon_i - 1))] \\ &= \sum [\ln(\epsilon_i - k_i) - \ln k_i - \ln(\epsilon_i - 1)] \end{aligned}$$

$$\sum g_0 I_{sp,i} \ln k_i - \Delta V = 0$$

We finally obtain the Lagrange function $\mathcal{L}$, which can then be used with the method already discuss to find a constrained maximum. We are searching for the best values for k_i that imply the highest possible ΔV .

$$\mathcal{L} = \sum [\ln(\epsilon_i - k_i) - \ln k_i - \ln(\epsilon_i - 1)] + \alpha \left(\sum c_i \ln k_i - \Delta V \right)$$

$$\begin{aligned} \frac{d\mathcal{L}}{dk_i} = 0 \quad \Rightarrow \quad & \sum \left(\frac{1}{\epsilon_i - k_i}(-1) - \frac{1}{k_i} \right) + \alpha \left(\sum c_i \frac{1}{k_i} \right) = 0 \\ & - \frac{1}{\epsilon_i - k_i} - \frac{1}{k_i} + \alpha \frac{c_i}{k_i} = 0 \\ & \frac{1}{\epsilon_i - k_i} = \frac{\alpha c_i}{k_i} - \frac{1}{k_i} \\ & (\epsilon_i - k_i)(\alpha c_i - 1) = k_i \\ & \alpha c_i \epsilon_i - \alpha c_i k_i - \epsilon_i + k_i = k_i \\ & k_i = \frac{\alpha c_i \epsilon_i - \epsilon_i}{\alpha c_i} = \frac{\epsilon_i(\alpha c_i - 1)}{\alpha c_i} \end{aligned}$$

These k_i correspond to the maximum ΔV . If we substitute them in the initial equation we can obtain α . Once found that value, we're able to substitute it back in the preceding formula and estimate at the end k_i .

$$\Rightarrow \alpha = 00.000543362 \quad \text{with an error 0.01}$$

$$k_1 = 2.700$$

$$k_2 = 3.315$$

$$k_3 = 2.896$$

	Tot/in	9.00	9.70	7.00				
	Isp	268.00	285.00	320.00				
	c	2629.08	2795.85	3139.2			DeltaV	9300
alpha		Kappa 1	Kappa 2	Kappa 3	DeltaV 1	DeltaV 2	DeltaV 3	Errore
0.00054336		2.700	3.315	2.896	2611	3351	3338	0.01
0.00055336		2.814	3.430	2.970	2720	3446	3418	283.64
0.00056336		2.924	3.542	3.042	2820	3536	3492	548.26
0.00057336		3.030	3.649	3.111	2914	3619	3563	795.86
0.00058336		3.132	3.753	3.178	3001	3697	3629	1028.14
0.00059336		3.231	3.853	3.242	3083	3771	3692	1246.58
0.00060336		3.326	3.950	3.304	3160	3841	3752	1452.45
0.00061336		3.419	4.044	3.365	3232	3906	3809	1646.87
0.00062336		3.508	4.134	3.423	3300	3968	3863	1830.83
0.00063336		3.595	4.222	3.479	3364	4027	3914	2005.19
0.00064336		3.679	4.307	3.534	3425	4083	3963	2170.73
0.00051640		2.371	2.982	2.682	2270	3054	3097	-879.19
0.00051740		2.384	2.994	2.690	2284	3066	3107	-843.12
0.00051840		2.397	3.007	2.699	2298	3078	3116	-807.34
0.00051940		2.409	3.020	2.707	2312	3090	3126	-771.86
0.00052040		2.422	3.033	2.715	2326	3102	3136	-736.66
0.00052140		2.435	3.046	2.723	2339	3114	3145	-701.76
0.00052240		2.447	3.059	2.731	2353	3126	3154	-667.13
0.00052340		2.460	3.071	2.740	2366	3137	3164	-632.78
0.00052440		2.472	3.084	2.748	2379	3149	3173	-598.70
0.00052540		2.484	3.097	2.756	2393	3160	3182	-564.90
0.00052640		2.497	3.109	2.764	2406	3171	3191	-531.36

FIGURE 2. Optimization process

LV Optimized Design				
Stage	1	2	3	Payload
Inert	10402	3094	931	2016
Prop	83215	26922	5588	
Total	93617	30016	6520	2016
Tot/in	9.00	9.70	7.00	
Initial	132169	38552	8535	2016
Final	48954	11630	2947	
Isp	268.0	285.0	320.0	
DeltaV	2611	3351	3338	9300
Kappa	2.700	3.315	2.896	
Lambda	0.292	0.221	0.236	

FIGURE 3. Optimized configuration

2.3 Optimized design

We now have $m_{u,3} = 2016 \text{ kg}$, when at the same time we're still guaranteeing $\Delta V = 9300 \text{ m/s}$. This opens up two choices: either, reduce the payload and get an higher ΔV or maintain the increased payload. We notice that it's convenient to move the mass from the third to the second stage.

- Configuration A
reduce the payload, obtain an higher ΔV

$$\Rightarrow \Delta V = 9314 \text{ m/s}$$

We gained 14.231 m/s .

LV initial configuration (a)						Sensitivity coefficients						(DeltaV Single Stage)			
Stage	1	2	3	Payload	%	Stage	1	2	3	Payload	Stage	1	2	3	
Inert	10402	3094	931	2000	0.0151	Inert	-0.0338	-0.2020	-0.9044	-0.9044	Inert	-0.0338	-0.1682	-0.7024	
Prop	83215	26922	5588		LV(excl PL)	Prop	0.0199	0.0387	0.1665		Prop	0.0199	0.0726	0.3685	
Total	93617	30016	6520	2000	130153	Total	0.0139	0.0139	0.0135		Total	0.0139	0.0477	0.2155	
Tot/in	9.00	9.70	7.00			Tot/in	0.621	0.768	1.425		Tot/in	0.621	0.768	1.425	
Initial	132153	38536	8520												
Final	48938	11614	2931												
Isp	268.0	285.0	320.0			Isp	9.745	11.766	10.466						
DeltaV	2612	3353	3349		9314										
Kappa	2.700	3.318	2.906												
Lambda	0.292	0.221	0.235												
	DeltaV nom		Variazione												
	9300		14.231												

FIGURE 4. Configuration A (max ΔV)

- Configuration B
maintain the optimized payload, check the ΔV

$$\Rightarrow \Delta V = 9314 \text{ m/s}$$

We gained 0.0118 m/s .

LV initial configuration (b)						Sensitivity coefficients					(DeltaV Single Stage)			
Stage	1	2	3	Payload	%	Stage	1	2	3	Payload	Stage	1	2	3
Inert	10402	3094	931	2016	0.0153	Inert	-0.0338	-0.2017	-0.8991	-0.8991	Inert	-0.0338	-0.1679	-0.6974
Prop	83215	26922	5588		LV(excl PL)	Prop	0.0199	0.0387	0.1661		Prop	0.0199	0.0725	0.3678
Total	93617	30016	6520	2016	130153	Total	0.0139	0.0139	0.0139		Total	0.0139	0.0477	0.2156
Tot/in	9.00	9.70	7.00			Tot/in	0.621	0.767	1.417		Tot/in	0.621	0.767	1.417
Initial	132169	38552	8535											
Final	48954	11630	2947											
Isp	268.0	285.0	320.0			Isp	9.743	11.757	10.432					
DeltaV	2611	3351	3338		9300									
Kappa	2.700	3.315	2.896											
Lambda	0.292	0.221	0.236											
	DeltaV nom		Variatione											
	9300		0.0118											

FIGURE 5. Configuration B (max $M_{u,3}$)

3 Part 3

3.1 Problem

Estimate the minimum payload that can be guaranteed assuming uncertainties on:

- 0.2% on the propellant mass $m_{p,1}$, $m_{p,2}$, $m_{p,3}$
- 3% on the inert mass of the solid stages $m_{s,1}$, $m_{s,2}$
- 5% on the inert mass of the liquid stage $m_{s,3}$
- 0.6% on the specific impulse of the solid stages $I_{sp,1}$, $I_{sp,2}$
- 1% on the specific impulse of the liquid stages $I_{sp,3}$

stage	m_s	m_p	I_{sp}
1st stage	3%	3%	5%
2nd stage	0.2%	0.2%	0.2%
3rd stage	0.6%	0.6%	1%

TABLE 2. Summary of the uncertainties

3.2 Sensitivity coefficients

From the final configuration, we are able to estimate the sensitivity coefficients, shown in Fig.6 in purple. These are coefficients that tell us how ΔV changes were to change for instance m_s or some other parameter. They are extremely important to understand how small changes or uncertainties, for the matter, can affect the final design. They are calculated as follows.

$$\begin{aligned}
 \frac{\partial \Delta V}{\partial I_{sp}} &= g_0 \ln \frac{m_0}{m_f} \\
 \frac{\partial \Delta V}{\partial m_p} &= g_0 I_{sp} \frac{1}{m_0} \\
 \frac{\partial \Delta V}{\partial m_s} &= \frac{\partial \Delta V}{\partial m_u} = g_0 I_{sp} \left(\frac{1}{m_0} - \frac{1}{m_f} \right) \\
 \frac{\partial \Delta V}{\partial m_{tot}} &= \frac{\partial \Delta V}{\partial m_s} \frac{1}{\epsilon} + \frac{\partial \Delta V}{\partial m_p} \frac{\epsilon - 1}{\epsilon} \\
 \frac{\partial \Delta V}{\partial \epsilon} &= \frac{m_s}{\epsilon} \left(\frac{\partial \Delta V}{\partial m_p} - \frac{\partial \Delta V}{\partial m_s} \right) \\
 \frac{\partial \Delta V_{tot}}{\partial m_{p,i}} &= \frac{\partial \Delta V_i}{\partial m_{p,i}} + \sum_{j=1}^{i-1} \frac{\partial \Delta V_j}{\partial m_{p,j}}
 \end{aligned}$$

LV configuration optimization						Sensitivity coefficients						(DeltaV Single Stage)					
Stage	1	2	3	Payload	%	Stage	1	2	3	Payload	%	Stage	1	2	3	Payload	%
Inert	10402	3094	931	2016	0.0153	Inert	-0.0338	-0.2017	-0.8991	-0.8991		Inert	-0.0338	-0.1679	-0.6974		
Prop	83215	26922	5588		LV(excl PL)	Prop	0.0199	0.0387	0.1661			Prop	0.0199	0.0725	0.3678		
Total	93617	30016	6520	2016	130153	Total	0.0139	0.0139	0.0139			Total	0.0139	0.0477	0.2156		
Tot/in	9.00	9.70	7.00			Tot/in	0.621	0.767	1.417			Tot/in	0.621	0.767	1.417		
Initial	132169	38552	8535														
Final	48954	11630	2947														
Isp	268.00	285.00	320.00			Isp	9.743	11.757	10.432								
DeltaV	2611	3351	3338		9300												
Kappa	2.700	3.315	2.896														
Lambda	0.292	0.221	0.236														
	DeltaV nom		Variatione														
	9300		0.012														

FIGURE 6. Final configuration

3.3 Effect of the uncertainties

These values, multiplied by the previous uncertainties entail a reduction of ΔV by 147 *m/s*. To remedy this inconvenience we need to reduce the payload instead. We're able to guarantee $m_{u,3} = 1852$ *kg*, that means we've lost 164 *kg* from the 2016 *kg* of the optimized design.

Design Uncertainties						DeltaV Uncertainties						Payload Uncertainties					
Stage	1	2	3			Stage	1	2	3								
Inert	3.00%	3.00%	5.00%			Inert	10.55	18.72	41.87	71.1448							
Prop	0.20%	0.20%	0.20%			Prop	3.31	2.08	1.86	7.2511							
Isp	0.60%	0.60%	1.00%			Isp	15.67	20.10	33.38	69.1528							
										147.5487		2016	-164	1852			

FIGURE 7. Sensitivities